

macro strategy selection model, etc. These parties have tried to market these services as an all-in-one algorithmic smart order solution. Without understanding the macro-level needs of the investor or their trading goals, it is simply not possible to provide this type of all-in-one solution. The best in class solutions to these algorithmic issues have followed our algorithmic decision making process: trading goal, adaptation tactic, limit order model, order submission rules, not the all-in-one type of solution.

ALGORITHMIC ANALYSIS TOOLS

Pre-Trade Analysis

The first step in developing an algorithmic trading strategy is to perform pre-trade analysis. This provides investors with the necessary data to make informed trading decisions on both the macro- and micro-levels, and serves as input into the algorithms. Pre-trade analysis provides investors with liquidity summaries, cost and risk estimates, and trading difficulty indicators as a means to screen which orders can be successfully implemented via algorithmic trading and which orders require manual intervention. It also provides potential risk reduction and hedging opportunities to further improve algorithmic execution strategies. Pre-trade data is comprised of current prices and quotes, liquidity and risk statistics, momentum, and an account of recent trading activity. This also provides investors with necessary data to develop short-term alpha models.

Intraday Analysis

Intraday analysis is used to monitor trading performance during trading. These systems will commonly provide in real-time the number of shares executed, the realized costs for those executed shares, the price movement since trading began (which translates to either a sunk cost or savings), and the expected market impact cost and timing risk for the remaining shares based on the implementation strategy and expected market conditions (which could be different from those expected at the beginning of trading). Some of the more advanced intraday analysis systems will provide z-score estimates, which are the projected risk adjusted trading costs for all shares (based on strategy and market conditions), as well as comparisons to projected final trading costs for various different algorithms and strategies. The intraday analysis systems are used by traders to evaluate market conditions and make revisions to their algorithms.